

LAPORAN PRAKTIKUM

STRUKTUR DATA

MODUL 2

STUKDAT



Disusun oleh:

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PROGRAM STUDI S1 INFORMATIKA

FAKULTAS INFORMATIKA

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LATIHAN SOAL MODUL

Link Repo Github Modul X : (<https://github.com/naufalakmall/struktur-data.git>)

1. Guided 1

Source Code

```
#include <iostream>
using namespace std;

int main() {

    int nilai1D[3] = {80, 85, 90};

    cout << "=== ARRAY 1 DIMENSI ===" << endl;

    cout << "Nilai pertama : " << nilai1D[0] << endl;
    cout << "Nilai kedua   : " << nilai1D[1] << endl;
    cout << "Nilai ketiga   : " << nilai1D[2] << endl;

    int nilai2D[2][3] = {
        {80, 85, 90},
        {75, 88, 92}
    };

    cout << endl;
    cout << "=== ARRAY 2 DIMENSI ===" << endl;

    cout << "Baris 0, Kolom 0 : " << nilai2D[0][0] << endl;
    cout << "Baris 0, Kolom 1 : " << nilai2D[0][1] << endl;
    cout << "Baris 1, Kolom 2 : " << nilai2D[1][2] << endl;

    int nilai3D[2][2][2] = {
        {
            {80, 85},
            {75, 90}
        },
        {
            {88, 92},
            {78, 86}
        }
    };

    cout << endl;
    cout << "=== ARRAY 3 DIMENSI ===" << endl;

    cout << "Data [0][0][0] : " << nilai3D[0][0][0] << endl;
```

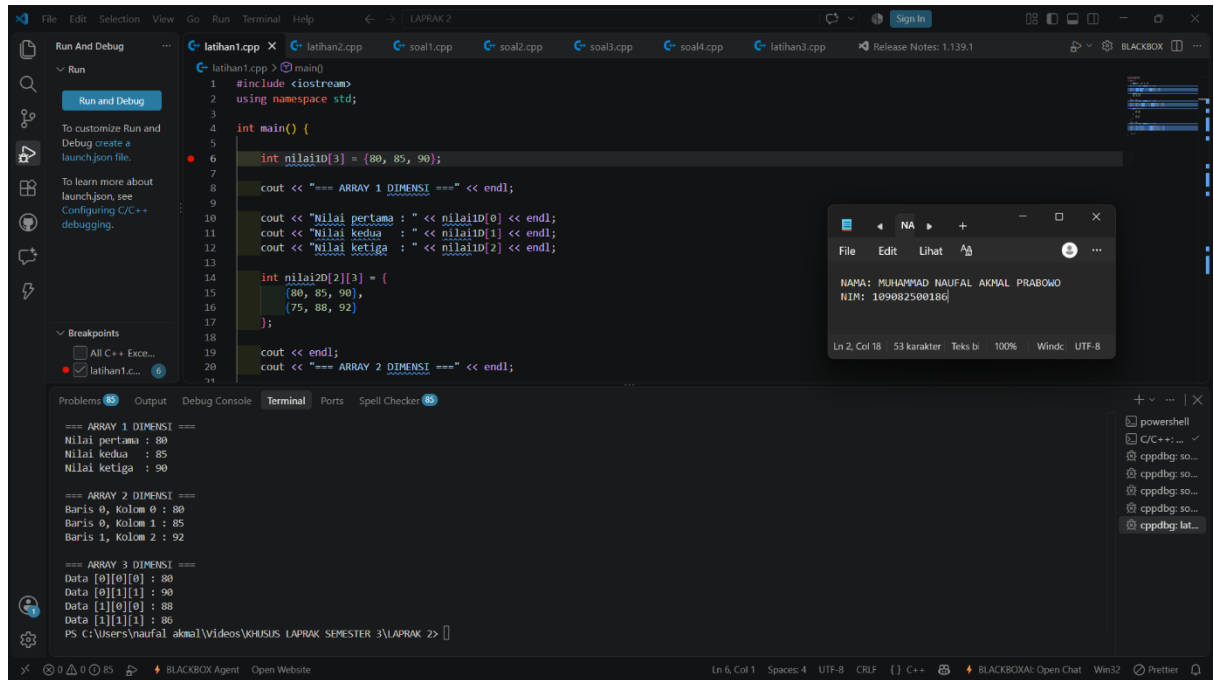
```

        cout << "Data [0][1][1] : " << nilai3D[0][1][1] << endl;
        cout << "Data [1][0][0] : " << nilai3D[1][0][0] << endl;
        cout << "Data [1][1][1] : " << nilai3D[1][1][1] << endl;

        return 0;
    }
}

```

Screenshoot program :



2. guided 2

Source Code

```
#include <iostream>
using namespace std ;

int maks3(int a, int b, int c) {
    int temp_max = a ;
    if (b > temp_max) {
        temp_max = b ;
    }
    if (c > temp_max) {
        temp_max = c ;
    }
    return temp_max ;
}

void tulis(int x) {
    for (int i = 0 ; i < x ; i++) {
        cout << "baris ke-" << i + 1 << endl ;
    }
}

int main() {

    int hasil_maks = maks3(10, 50, 30) ;
    cout << "Nilai maksimumnya adalah = " << hasil_maks <<
endl ;

    cout << "Mulai panggil prosedur: " << endl ;
    tulis(3) ;

    return 0 ;
}

    "lima", "enam", "tujuh", "delapan", "sembilan"
};

    if (angka < 10) {
        cout << angka << " : " << satuan[angka];
    }
    else if (angka < 20) {
        if (angka == 10)
            cout << angka << " : sepuluh";
        else if (angka == 11)
            cout << angka << " : sebelas";
        else
            cout << angka << " : " << satuan[angka-10] <<
" belas";
    }
    else {
```

```

int puluhan = angka / 10;
int sisa = angka % 10;

string puluh[] = {
    "", "", "dua puluh", "tiga puluh",
    "empat puluh", "lima puluh",
    "enam puluh", "tujuh puluh",
    "delapan puluh", "sembilan puluh"
};

cout << angka << " : " << puluh[puluhan];

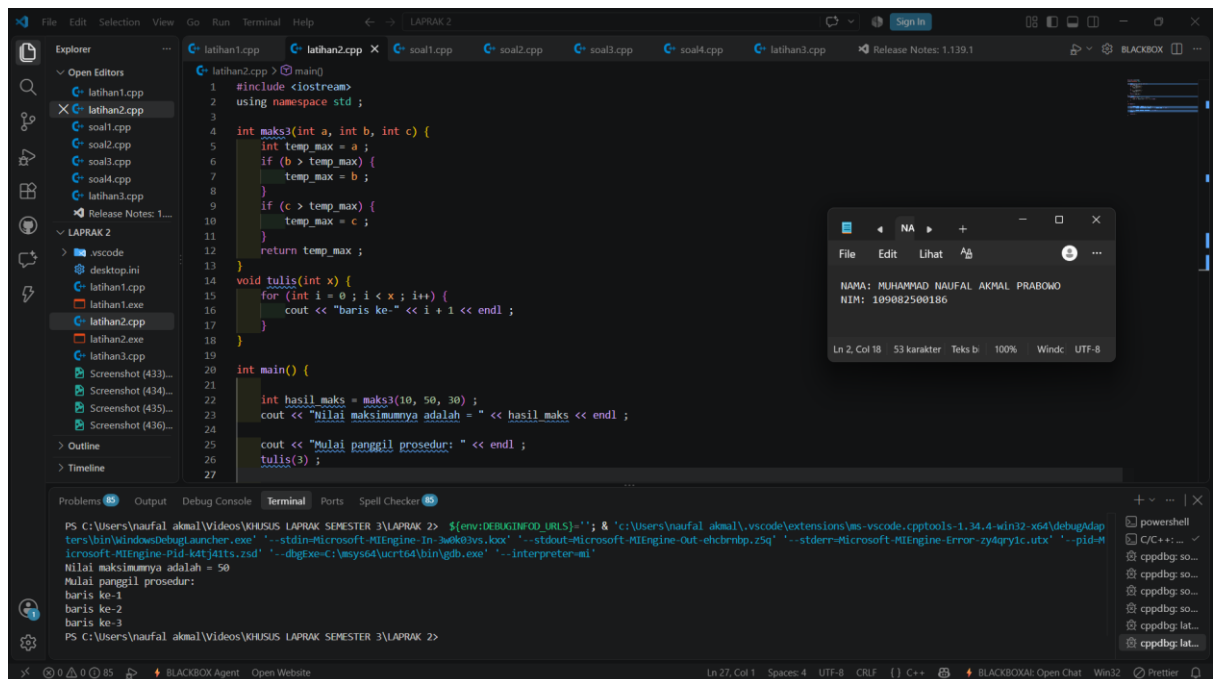
if (sisa != 0) {
    cout << " " << satuan[sisa];
}

}

return 0;
}

```

Screenshoot program



3. Guided 3

Source Code

```
#include <iostream>
using namespace std ;

void tukarValue(int x, int y) {
    int temp = x ;
    x = y ;
    y = temp ;
}

void tukarPointer(int *px, int *py) {
    int temp = *px ;
    *px = *py ;
    *py = temp ;
}

void tukarReference(int &px, int &py) {
    int temp = px ;
    px = py ;
    py = temp ;
}

int main() {
    int a = 4, b = 6 ;

    cout << "Kondisi Awal -> a: " << a << " b: " << b << endl
    ;

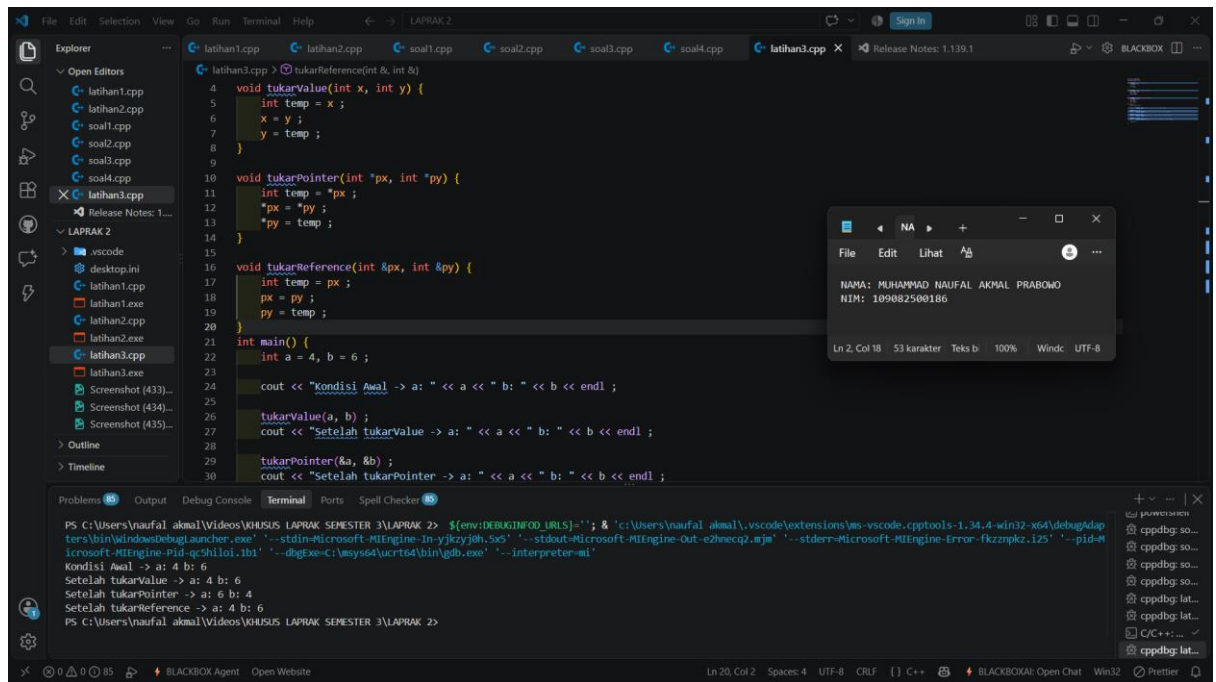
    tukarValue(a, b) ;
    cout << "Setelah tukarValue -> a: " << a << " b: " << b
    << endl ;

    tukarPointer(&a, &b) ;
    cout << "Setelah tukarPointer -> a: " << a << " b: " << b
    << endl ;

    tukarReference(a, b) ;
    cout << "Setelah tukarReference -> a: " << a << " b: " <<
    b << endl ;

    return 0 ;
}
```

Screenshoot program



4. Soal 1

Source Code

```
#include <iostream>
#include <vector>

using namespace std;

int main() {
    int n;
    cin >> n;

    vector<int> nilai(n);
    int total = 0;

    for (int i = 0; i < n; i++) {
        cin >> nilai[i];
        total += nilai[i];
    }

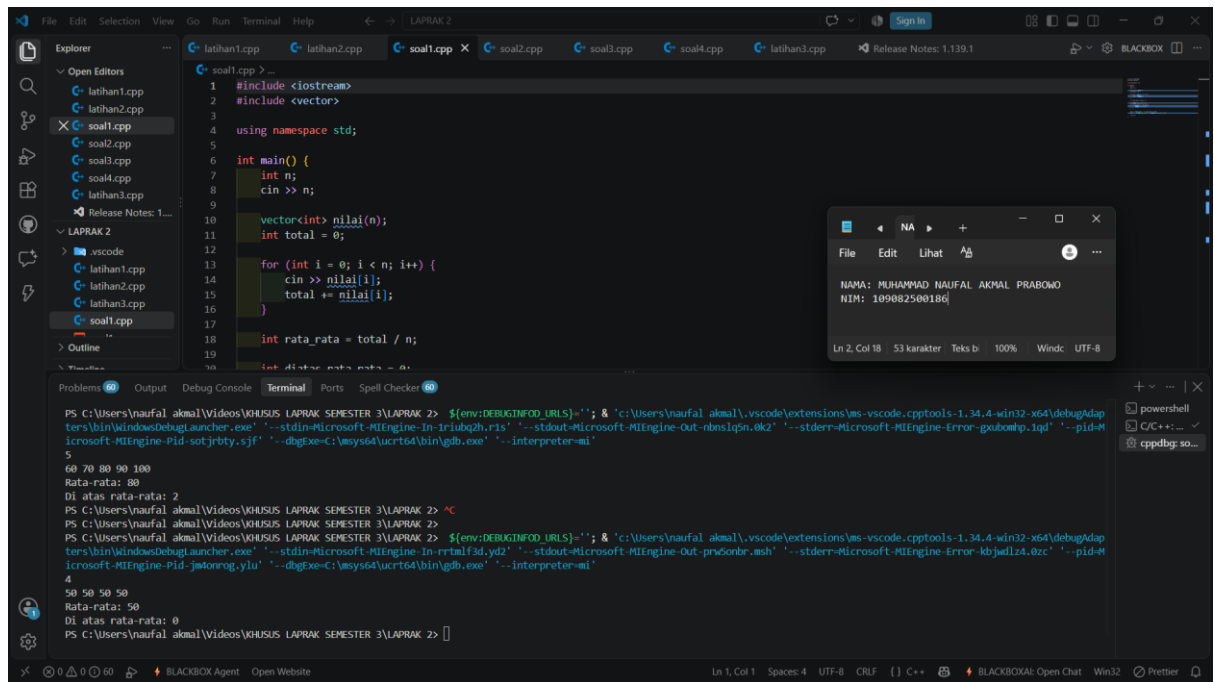
    int rata_rata = total / n;

    int diatas_rata_rata = 0;
    for (int i = 0; i < n; i++) {
        if (nilai[i] > rata_rata) {
            diatas_rata_rata++;
        }
    }

    cout << "Rata-rata: " << rata_rata << endl;
    cout << "Di atas rata-rata: " << diatas_rata_rata <<
endl;

    return 0;
}
```

Screenshoot program



Deskripsi Program :

Program C++ ini menghitung nilai rata-rata dari sekelompok data nilai mahasiswa yang disimpan dalam array 1 dimensi dan menentukan berapa banyak mahasiswa yang memperoleh nilai di atas rata-rata tersebut.

5. Soal 2

Source Code

```

#include <iostream>

using namespace std;

int main() {
    int matriks[3][3];
    int jumlah_diagonal = 0;

    for (int i = 0; i < 3; i++) {
        for (int j = 0; j < 3; j++) {
            cin >> matriks[i][j];
        }
    }

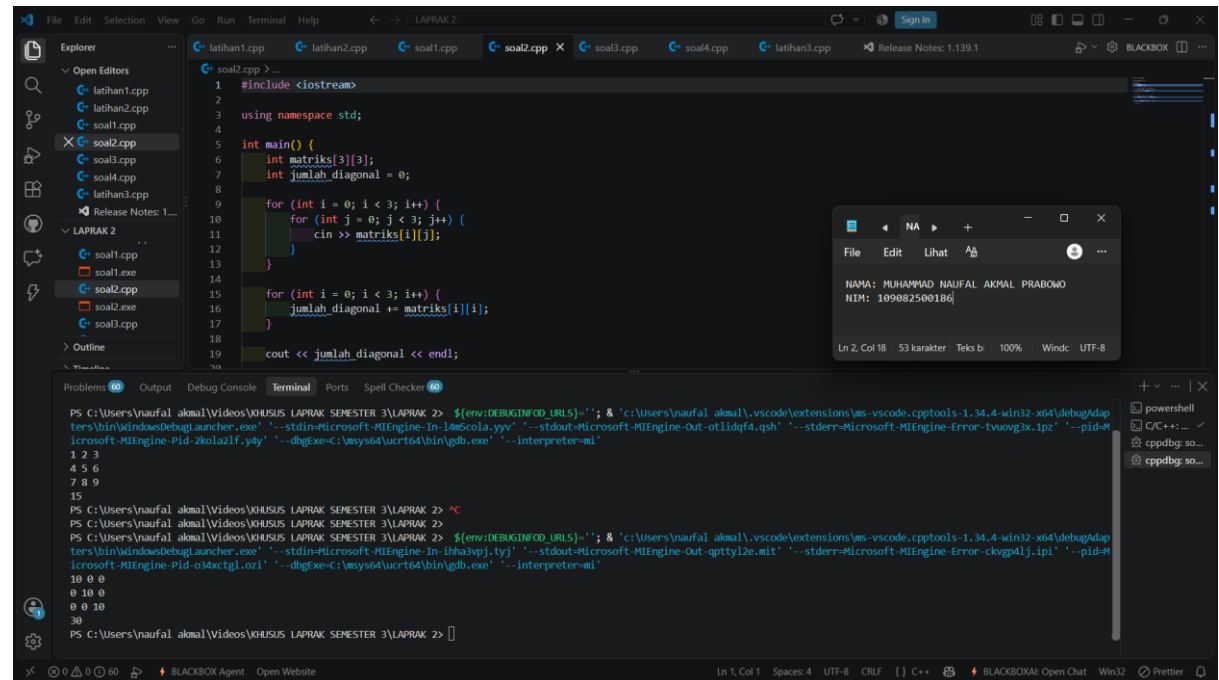
    for (int i = 0; i < 3; i++) {
        jumlah_diagonal += matriks[i][i];
    }

    cout << jumlah_diagonal << endl;

    return 0;
}

```

Screenshoot program



Deskripsi Program :

Program C++ ini mengimplementasikan sebuah fungsi untuk menghitung dan mengembalikan jumlah kemunculan suatu karakter tertentu di dalam sebuah string.

6. Soal 3

Source Code

```
#include <iostream>
#include <string>

using namespace std;

int hitungKarakter(const string& kata, char target) {
    int jumlah = 0;
    for (char c : kata) {
        if (c == target) {
            jumlah++;
        }
    }
    return jumlah;
}

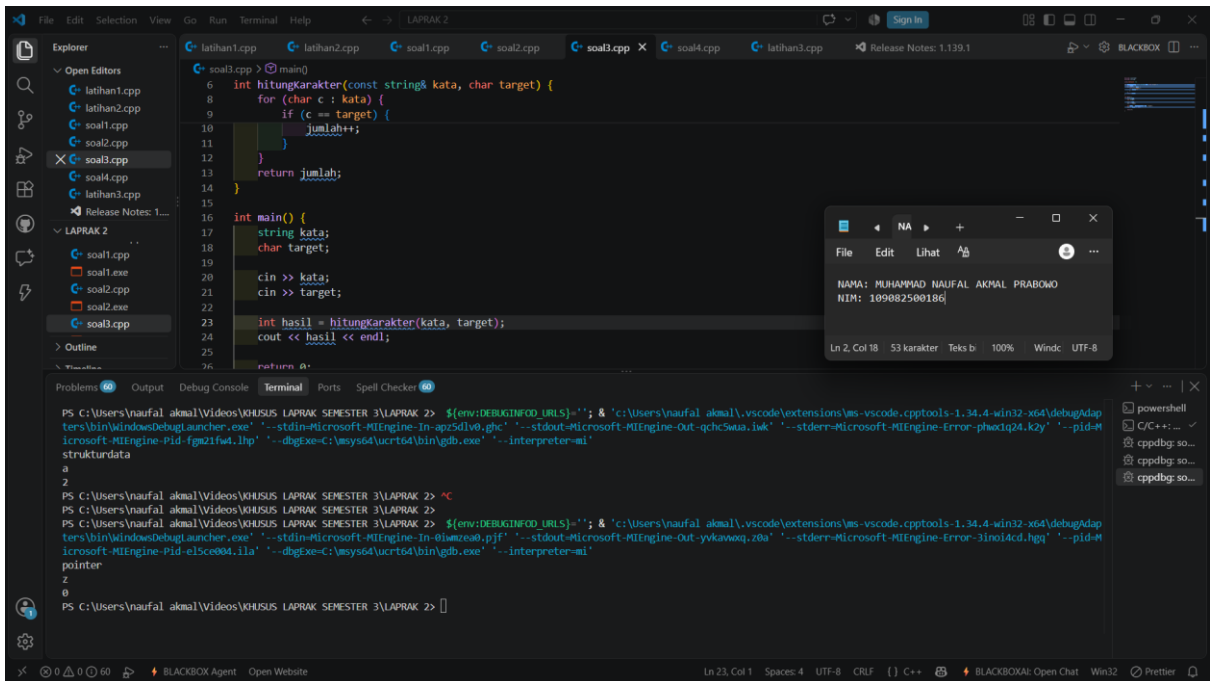
int main() {
    string kata;
    char target;

    cin >> kata;
    cin >> target;
```

```
int hasil = hitungKarakter(kata, target);
cout << hasil << endl;

return 0;
}
```

Screenshot program



Deskripsi Progam :

Program C++ ini mengimplementasikan sebuah fungsi untuk menghitung dan mengembalikan jumlah kemunculan suatu karakter tertentu di dalam sebuah string.

7. Soal 4

Source Code

```
#include <iostream>

using namespace std;

void tukarDanKalikan(int &a, int &b) {
    int temp = a;
    a = b;
    b = temp;

    a *= 10;
    b *= 10;
}

int main() {
    int x, y;
```

```

    cin >> x >> y;

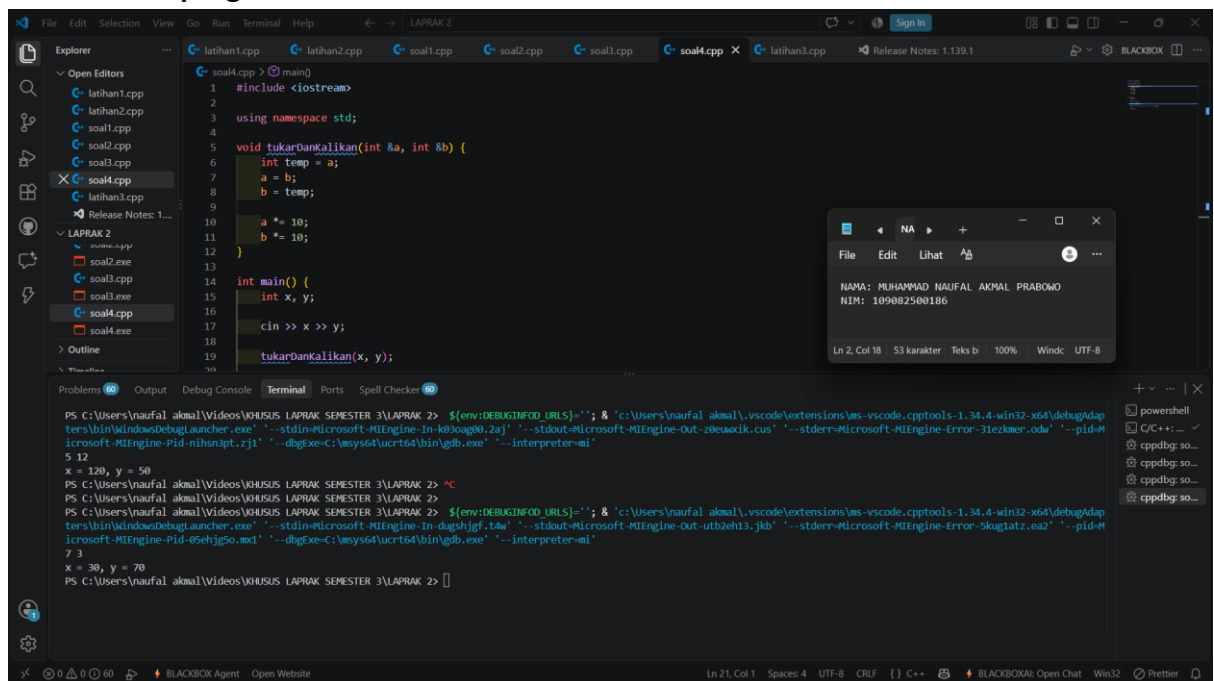
    tukarDanKalikan(x, y);

    cout << "x = " << x << ", y = " << y << endl;

    return 0;
}

```

Screenshoot program



Deskripsi Program :

Program C++ ini mengimplementasikan sebuah prosedur dengan parameter pass by reference untuk menukar nilai dari dua variabel bulat lalu mengalikan masing-masing hasilnya dengan angka 10.